

INTRODUCTION

Chromosome 22 contains many genes that are linked to various biological functions and diseases. In this project, we analyzed Carl Zimmer's genome to identify the top genes in chromosome 22 based on mutational burden, i.e. number of single nucleotide polymorphisms (SNPs), and impact score. Using the Genotype-Tissue Expression (GTEx) Project database, we link the expression of these genes to their reported functions in literature and highlight the potential relevance of variants to disease states.

METHODS

We first sought to identify the top 10 genes with the highest mutational load on chromosome 22. To do this, we used the `cyvcf2` library in Python to read in Carl Zimmer's germline SNP call set VCF file and convert it into a dataframe. We then filtered the dataset to retain only those variants located on chromosome 22. As all variants had passed filtering and the lowest quality score observed was 30.74, we considered this set of variants to be of sufficient quality for downstream analysis.

To obtain gene-level annotations, we annotated the viable variants using the Ensembl Variant Effect Predictor (VEP, v113.0) with the following configuration: GRCh38.p14 assembly, Ensembl transcripts as the reference transcript database, and various additional annotations enabled including 1000 Genomes global minor allele frequency, gene symbols, APPRIS, MANE, SIFT and PolyPhen predictions and scores, regulatory region consequences, GENCODE version 47 as the gene model, dbSNP build 156 for known variant reference, and PubMed IDs for co-located variants. The upstream and downstream distance for regulatory annotation was set to 5000 base pairs.

Following annotation, we grouped the variants by gene and counted the number of variants associated with each gene. We then ranked the genes based on the number of variants, identifying the top 10 genes with the highest mutational burden on chromosome 22. These genes were subsequently selected for further study using the GTEx database, which provides protein expression levels in different tissues. Based on which tissues exhibit highest expression levels of each protein, the inferred functions of these genes were linked to their known roles in various biological pathways and diseases through literature review.

Using the annotated file containing only variants on chromosome 22, we looked to prioritize genes through other means. Initially, we had attempted to use clinical significance to find a new top 10 genes. However, it turned out that there were only eight variants that held clinical significance, not enough to make a new list of prioritized genes.

Since clinical significance was not a viable option, we decided to turn to impact. In Ensembl, impact is the predicted severity of the variant depending on the mutation. For example, a mutation in a non-coding region would be listed as “MODIFIER”, indicating that prediction is difficult or there is no evidence of impact. On the other end of the spectrum, a mutation that results in early truncation would be listed as “HIGH”, as a truncated protein would likely have a strong effect on the protein’s function. The possible values for impact include “MODIFIER”, “LOW”, “MODERATE”, and “HIGH”, going from lowest to highest predicted consequence. By sorting the genes via impact, we would be able to find variants that would likely have an effect in coding regions, filtering out genes that would likely have no effect on the individual.

In order to prioritize a new set of 10 genes according to impact, we calculated the total impact found in a gene with a weighted scoring system. Here, we had “MODIFIER” = 0, “LOW” = 1, “MODERATE” = 2, and “HIGH” = 3. Of note, “MODIFIER” = 0 because every variant had an impact score, many of which were “MODIFIER”. If “MODIFIER” were to have any value other than zero, then it would most likely overshadow variants that had impact scores other than “MODIFIER”. Once the impact totals had been calculated, we received a new top 10 genes. Of these genes, only CACNG2 and EFCAB6 were found in both lists.

RESULTS

Analysis of the GTEx expression profiles and literature review of the top 10 genes identified yielded the following results:

LARGE1’s role in muscle tissue aligns with its GTEx expression profile, which shows highest expression in the bladder, sigmoid colon, left ventricle, and uterus. These are tissues that all contain high proportions of muscle tissue. LARGE1 is a glycotransferase that generates matriglycan, which is polysaccharide that connects the cytoskeleton and extracellular matrix, and is required for proper muscle function (Katz, 2022). Mutations in LARGE1 have been linked to congenital muscular dystrophy and can present with brain abnormalities, which can also be connected to increased expression levels in certain brain tissues (Joseph et al., 2022).

TAF5 has relatively high expression in all brain tissues, which corresponds to the fact that it is found in areas of inflammatory activation in the brain, triggering production of astrocytes after CNS injury (Huang et al., 2020). It is also produced during neurogenesis of neural stem cells (Shahapal et al., 2019). Genetic depletion of Tafa5 results in depressive behaviors and decreased hippocampus activity, and chronic stress-treated mice had decreased plasma levels of this protein, demonstrating TAF5’s role in mood and cognition (Huang et al., 2020). Curiously, there is also significant expression in the tibial nerve, which has not been associated with any specific function thus far.

SYN3 has the highest expression in all brain tissues, most notably the cerebral cortex. This reflects its function since this protein, as part of the synapsin family, is a neuronal

phosphoprotein which plays a significant role in synaptic transmission, neuronal plasticity, axon growth, and synapse formation (Cruceanu et al., 2013). Unlike the other synapsins which are most prominently located in nerve terminals, SYN3 immunostaining of adult mouse brains shows they are localized to brain regions that are undergoing high levels of adult neurogenesis. Genetic variants for this gene have been associated with various brain disorders, such as epilepsy, schizophrenia, and bipolar disorder.

TBC1D22A has broad expression among all tissue types, with highest levels in EBV-transformed lymphocytes, spleen, and the thyroid. This suggests it may play a role in the immune system. Its known function is as a GTPase activating protein of Rab40 which regulates lipophagy, and loss of TBC1D22A results in lipid droplet accumulation across various tissues (Duan et al., 2021). Missense mutations in this gene have also been correlated with platinum-sensitivity in ovarian cancer patients (Lv et al., 2024).

MIATNB shows highest expression in skin, blood, and spleen, but low expression in brain tissues, suggesting tissue-specific regulatory roles in epithelial integrity and immune response. This implies potential roles in inflammation, immunity. MIATNB regulates autophagy and apoptosis in corneal cells under hyperosmotic stress, suggesting its therapeutic potential in dry eye disease (Pan et al., 2023). MIATNB is also associated with Glioma Susceptibility 1 (Weizmann Institute of Science, n.d.).

MIR3667HG is most highly expressed in spleen, lung, and EBV-transformed lymphocytes. In literature, it is enriched in extracellular vesicle DNA from pancreatic cancer cells, showing cancer-specific coding DNA signatures and potential as a biomarker distinguishing malignant from non-cancerous cell types (Olou et al., 2025). Diseases associated with this gene include Glioma Susceptibility 1 and Immunodeficiency-Centromeric Instability-Facial Anomalies Syndrome 1 (GeneCards, n.d.)

The CACNG2 gene is solely expressed in brain tissues, and encodes the brain-specific calcium channel subunit $\gamma 2$ (stargazin), vital for AMPA receptor trafficking and synaptic plasticity. One study links CACNG2 SNPs to neuropathic pain risk (Mejía-Terrazas et al., 2022), while another shows its transcription is regulated by the cAMP–PKA–CREB pathway via a lncRNA (Muñoz-Herrera et al., 2023).

TTC28 is primarily expressed in the female reproductive system, including the vagina, ovaries, and uterus, but it is expressed in many other tissues and organs as well. Mutations in TTC28 and downregulation are linked to several types of cancer, as it has a large role in regulating mitosis and cytokinesis (Zhang et al., 2024). It also contributes to chromosomal stability, meaning that its downregulation leads to the decreased chromosome stability of cancerous cells (Zhang et al., 2025).

EFCAB6 is expressed primarily in the testes. Mutations in EFCAB6 have been shown to decrease sperm motility and fertility in mice. EFCAB6 controls the activity of CatSper channel

complexes, which is a calcium channel that responds to changes in pH in vaginal fluid (Hwang et al., 2019). Additionally, in males with CAIS (having XY chromosomes, but developing female genitalia), EFCAB6 is not expressed (Malcher et al., 2019).

LINC02885, also referred to as ISX-AS1, is a long non-coding RNA most highly expressed in the brain, particularly in the hypothalamus, frontal cortex, and hippocampus. In patients with central nervous system tumors such as lower grade gliomas, LINC02885 has been linked with higher overall survival rates, suggesting a protective role against them (Śledzińska et al., 2021). It has also been found to be upregulated in individuals with gastric cancer that experience peritoneal metastasis (Lee et al., 2024).

DISCUSSION

The results signified a diverse set of genes including genes involved in protein coding and non coding RNAs which were slightly less distinguished. Some genes like **CACNG2** and **EFCAB6** were noticed in both analysis of mutational burden and variant impact based approaches which further strengthened their relevance. The tissue specific analysis also provided some additional insights into potential roles of our shortlisted genes in health and disease. Multiple genes were greatly expressed in immune, muscle, reproductive and brain tissues indicating their favorability to neural function, reproductive biology, muscle regulations and immunity. The patterns also suggested how their dysregulations lead to various ailments. Genes that were highly expressed in certain tissue types are most likely to play a crucial role in sustaining the specialized function of those tissues. For example, those with a high expression in immune or neural tissues could be implicated in inflammatory or neurological disorders while those enriched in reproductive organs could influence fertility or reproductive health. Furthermore, theories on underlying regulatory mechanisms, such as epigenetic control or localized transcription factor can be guided by tissue-specific expression. These results underscore the need to adapt functional research and treatment development initiatives to the specific tissue setting but overall prioritizing targets in genome research has been made easier by knowing where and how these genes are expressed, particularly for precision medicine approaches that look forward to creating tissue-selective biomarkers for a more personalized treatment. Future work could include performing in-depth analysis of the specific SNPs identified in Carl Zimmer's genome and comparing those to known disease-causing mutations or variants reported in the literature. Additionally, protein structure modelling of Zimmer's SNPs could reveal if any specific SNPs are predicted to have significant impacts on the proper folding, and thus function, of these proteins.

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